



Relative efficacy of antianginal drugs used as add-on therapy in patients with stable angina: A systematic review and meta-analysis

Jonathan Belsey¹, Irene Savelieva², Alessandro Mugelli³ and A John Camm²

Abstract

Aims: First-line medical management of stable angina generally involves a beta-blocker (BB) or calcium channel blocker (CCB), with other classes of medication being added if symptom control is inadequate. Evidence supporting the appropriate choice of a second-line agent is currently unclear. The objective of this systematic review was to quantify the clinical benefit of BB, CCBs, long-acting nitrates (LANs), ranolazine, trimetazidine, ivabradine or nicorandil added to first-line monotherapy for stable coronary artery disease.

Methods: Randomised controlled trials comparing the efficacy of antianginal therapies in patients with stable angina refractory to first-line therapy were identified from a literature search. Exercise tolerance test (ETT) data and clinical outcomes were extracted and combined in a series of meta-analyses.

Results: A total of 46 qualifying studies were identified, evaluating 71 treatment comparisons. The combination of ranolazine added to CCB or BB showed positive outcomes across all outcomes assessed. Other combinations of BB, CCB, LAN and trimetazidine showed significant benefits for most but not all outcomes. Ivabradine demonstrated benefits for ETT assessments but these were not matched in clinical domains. No qualifying studies were identified for nicorandil in an add-on role.

Conclusion: Across a range of commonly assessed exercise and clinical outcomes, the effectiveness of BB+CCB used in combination is broadly confirmed. Ranolazine used with BB or CCB showed benefits across all outcomes assessed, while LAN and trimetazidine used with BB or CCB have shown benefits across some outcomes. Ivabradine added to BB shows inconsistent effects from a single study, whilst there is no relevant evidence for nicorandil.

Keywords

Stable angina, antianginal therapy, meta-analysis, beta blocker, calcium antagonist, trimetazidine, nicorandil, ranolazine, ivabradine long acting nitrate

Received 17 February 2014; accepted 5 April 2014

Introduction

Published clinical guidelines on the management of stable angina^{1–3} differ in the details of recommended care pathways for individual patients, reflecting cultural, regulatory organizational and financial differences across Europe and North America. Regarding symptomatic pharmacological management, there is considerable agreement. Generally, if intermittent use of short-acting nitrates proves insufficient to control symptoms, then regular prophylaxis using a beta-blocker (BB) should be started. Where this is

contraindicated or proves poorly tolerated, a calcium channel blocker (CCB) should be used instead. Whilst neither of these treatments has been shown to improve outcomes in patients with chronic stable angina, their

¹JB Medical Ltd, UK

²St George's, University of London, UK

³University of Florence, Italy

Corresponding author:

Jonathan Belsey, JB Medical Ltd, The Old Brickworks, Chapel Lane, Little Cornard, Sudbury, CO10 0PB, UK.
Email: jbelsey@jbmedical.com

first-line use is supported by a wide range of evidence and clinical experience.

Where monotherapy with BB or CCB fails to control symptoms, treatment choice tends to be more empirically based. The 2011 guidelines published by the UK National Institute for Health and Clinical Excellence (NICE) suggest that, when a patient experiences residual anginal symptoms on either a BB or CCB, agents from both classes should be prescribed together.¹ Where this option fails to control symptoms, NICE recommends that one of the first-line agents should be stopped, and an alternative agent be prescribed, with the selection being made on an individual basis, based on consideration of co-morbidities, contra-indications, patient preference and drug costs. The 2013 European Society of Cardiology (ESC) Guidelines on the management of stable coronary artery disease (SCAD) advocate the use of long-acting nitrates (LANs), ivabradine, nicorandil, or ranolazine (class of recommendation IIa, level of evidence (LOE) B) or trimetazidine (IIb, LOE B) as second-line therapy according to individual patient characteristics such as heart rate, blood pressure and drug tolerance.²

This advice reflects limited published evidence for add-on therapy with older anti-anginals. There are a few large prospective studies relating to the use of ranolazine and ivabradine, but far less impressive evidence regarding other treatment choices. A previous systematic review of studies published up to 1999⁴ identified 22 small studies of variable quality that evaluated the exercise tolerance test (ETT) impact of add-on BB and CCB, but no evidence was offered for other treatment choices or clinical outcomes (angina frequency and nitrate use). Since 1999, a number of alternative options have become available and a considerable number of relevant clinical trials have been published.

The objective of this systematic review was to update and extend the previously published review, based on published clinical trials investigating the use of BB, CCB, LAN, nicorandil, ranolazine, trimetazidine or ivabradine added in to either BB or CCB as first-line agents. Based on this systematic review, the intention was to extract data on both ETT and clinical outcomes, and to carry out meta-analyses in order to inform prescribing choice.

Methods

Literature review

Searches were carried out on MEDLINE, EMBASE and the Cochrane Central Register of Clinical Trials up to September 2013 to identify randomised controlled trials of anti-anginal drugs used as add-on therapy to either BB or CCB for SCAD. The following search terms were used for the MEDLINE search,

with equivalent terms being adopted for EMBASE and Cochrane (MeSH = Medical Subject Heading, PT = Publication Type, TW = Text Word): Angina pectoris [MeSH] AND Randomized controlled trial [PT] OR Randomised [TW] OR Randomized [TW] AND Adrenergic beta antagonists [MeSH] OR Calcium channel blockers [MeSH] OR Nitrates [MeSH] OR Nicorandil [MeSH] OR Ranolazine [TW] OR Trimetazidine [TW] OR Ivabradine [TW].

Identified abstracts were scrutinised to ascertain whether they fulfilled the inclusion criteria: randomised controlled trial; parallel group or crossover design; at least two treatment groups investigating first-line therapy alone versus first-line plus an agent from another class; patients to have symptomatic stable angina; each treatment to be evaluated for a minimum of two weeks, with at least one week washout between treatments in the case of crossover studies; sufficient data reported to allow data on one or more of the following outcomes to be extracted: (a) ETT – total exercise duration, (b) ETT – time to 1 mm ST depression, (c) ETT – time to onset of angina, (d) weekly angina attack frequency, (e) weekly use of short-acting nitrates.

Where abstracts indicated that a study was likely to fulfil these criteria, the full text was retrieved. Reference lists of these full texts were inspected to identify studies that had not been identified in the primary searches. Additional searches were carried out to identify previous systematic reviews in the field and the reference lists examined to identify any additional studies.

Data extraction

Studies included in the analysis were grouped according to the treatment comparisons. CCB added to BB and BB added to CCB were separately grouped. For the other therapeutic classes, studies investigating either BB or CCB as baseline therapy were grouped together.

Exercise tolerance test

Duration of exercise tolerance ($\pm$ standard deviation) at the end of each treatment period, for each of the three exercise endpoints specified above were extracted. Where measurements were available at both peak and trough drug levels, the trough level results were used. No exclusions were made on the grounds of the exercise protocols used, as the non-parametric meta-analytical method used allows results of dissimilar tests to be compared meaningfully (see statistical analysis below).

Attack frequency and nitrate use

Data on the frequency of angina attacks and the number of supplementary short-acting nitrate doses

used were extracted. The data were adjusted to yield a weekly rate for both outcomes. Where data were only presented in graphical format, estimates of the values were extracted by enlargement and scaled measurement. Where data were presented as median and range only, the mean and standard deviation were estimated using the method of Hozo et al.⁵

Statistical analysis

Results for each combination of first-line and add-on therapy were pooled using a random effects model to arrive at an overall estimate of weighted relative benefit for each treatment class.⁶ Random effects is a more conservative method than fixed effects and, although it yields broader confidence intervals, it is the method of choice where heterogeneity is likely. Where heterogeneity does not exist, it will yield identical results to a fixed effects model.

For the ETT data, the standardised mean difference (SMD) was the measure of choice. The SMD is estimated by a weighted pooling of the mean difference in exercise duration for each study, divided by the standard deviation. The SMD has the important property that its value does not depend on the measurement scale used in the study. Thus, if several trials are identified assessing the same outcome, but using different scales or protocols, the standardised mean difference will convert all outcomes to a common scale, measured in units of standard deviations. Although the measure itself is dimensionless, it allows meaningful quantitative comparisons between studies to be made regardless of the ETT protocol used.

For the clinical outcomes, a relative rate reduction in weekly angina rate or short acting nitrate use was measured instead, as there is no issue of varying scales to consider for these outcomes. This was calculated as the mean difference in treated attack rate or nitrate use between monotherapy and dual therapy groups, divided by the rate seen in the monotherapy control group, expressed as a rate ratio.

In both cases, the denominator used was the intention to treat population.

Subgroup analyses

Exploratory analyses were carried out for the following subgroups of studies: (1) limiting the analysis set to studies published before or after 1990; (2) subdividing the analysis set into studies with low and high baseline angina frequency, based on a threshold of seven attacks per week; (3) subdividing the analysis set into studies with more or less advanced disease, based on the presence of one or more of the following characteristics: (a) >75% of patients with multivessel disease; (b) >10% of patients with clinical heart failure (New York Heart

Association (NYHA) grade III or IV), and (c) >25% of patients with prior revascularisation. Thresholds were set post hoc, based on an assessment of the distribution of relevant characteristics in the dataset.

Results

After carrying out the specified searches and assessing for qualification to the inclusion criteria, we identified 46 qualifying studies, comprising 71 relevant treatment comparisons (see online Supplementary Material, Figure S1).

There were 28 studies (37 comparisons) of a CCB added to a BB ($n=2468$);^{7–35} 12 studies (12 comparisons) of a BB added to a CCB ($n=1489$);^{9,11,12,14–16,19,20,24,25,29,34} 7 studies (8 comparisons) of a LAN added to a BB or CCB ($n=329$);^{7,36–40} 3 studies (4 comparisons) of ranolazine added to a BB or CCB ($n=1388$);^{41–43} 8 studies (9 comparisons) of trimetazidine added to a BB or CCB ($n=2875$);^{44–51} 1 study of ivabradine added to a BB ($n=889$).⁵² No studies were found that evaluated the impact of nicorandil in an add-in role on either ETT or clinical measures of anti-anginal efficacy. Full details of study types, sizes and treatments are provided in online Supplementary Material, Table S1.

Results for all outcomes, broken down by class are presented in Table 1 and Figure 1. Forest plots for all outcomes are provided in online Supplementary Material, Figures S1–S5.

Effect of treatment on exercise tolerance test

For total exercise time, statistically significant benefits were seen for ranolazine added to BB or CCB, LAN added to BB or CCB, CCB added to BB, ivabradine added to BB (single study), BB added to CCB and trimetazidine added to BB or CCB (online Supplementary Material, Figure S2). For time to ST segment depression, statistically significant benefits were seen for LAN added to BB or CCB, ranolazine added to BB or CCB, ivabradine added to BB (single study) and trimetazidine added to BB or CCB.

There was no significant difference seen for CCB added to BB or for BB added to CCB. For time to angina, statistically significant benefits were seen for CCB added to BB, ranolazine added to BB or CCB, BB added to CCB, ivabradine added to BB (single study) and trimetazidine added to BB or CCB. There was no significant difference seen for LAN added to BB or CCB.

Effect of treatment on attack frequency and nitrate use

For weekly angina frequency, statistically significant benefits were seen for ranolazine added to BB or

Table 1. Summary of results of meta-analyses for each class of anti-anginal drugs for the five outcomes explored.

Baseline therapy Date range	Add-on therapy	ETT outcomes (standardised mean difference $\pm$ 95% CI)			Clinical outcomes (relative rate reduction $\pm$ 95% CI)		
		Total time	Time to ST depression	Time to angina	Angina frequency	Nitrate use	
BB 1981–2000	CCB	0.12 (0.07–0.17) $p < 0.0001$ N = 2571	0.20 (0.00–0.40) $p = NS$ N = 2207	0.30 (0.22–0.38) $p < 0.0001$ N = 2210	0.21 (0.08–0.32) $p = 0.002$ N = 2149	0.23 (0.06–0.36) $p = 0.01$ N = 1624	
CCB 1983–2000	BB	0.09 (0.01–0.17) $p = 0.04$ N = 986	0.22 (–0.01–0.45) $p = NS$ N = 1075	0.20 (0.10–0.31) $p = 0.0002$ N = 1007	0.30 (–0.55–0.68) $p = NS$ N = 921	0.33 (0.21–0.43) $p < 0.0001$ N = 800	
BB or CCB 1983–1991	LAN	0.14 (0.00–0.28) $p = 0.04$ N = 332	0.29 (0.13–0.44) $p = 0.0004$ N = 200	0.08 (–0.06–0.23) $p = NS$ N = 332	0.19 (0.11–0.25) $p < 0.0001$ N = 296	0.13 (0.04–0.20) $p = 0.004$ N = 188	
BB or CCB 2004–2013	Ranolazine	0.20 (0.19–0.21) $p < 0.0001$ N = 1092	0.18 (0.14–0.23) $p < 0.0001$ N = 1092	0.23 (0.21–0.25) $p < 0.0001$ N = 1092	0.22 (0.10–0.32) $p = 0.0006$ N = 2040	0.32 (0.22–0.40) $p < 0.0001$ N = 2040	
BB or CCB 1984–2011	Trimetazidine	0.06 (0.02–0.11) $p = 0.006$ N = 3493	0.12 (0.05–0.19) $p = 0.001$ N = 3493	0.17 (0.10–0.25) $p < 0.0001$ N = 3203	0.21 (0.01–0.36) $p = 0.04$ N = 3224	0.21 (–0.02–0.38) $p = NS$ N = 3238	
BB 2009	Ivabradine	0.10 ^a (0.09–0.11) $p < 0.0001$ N = 889	0.17 ^a (0.16–0.18) $p < 0.0001$ N = 889	0.18 ^a (0.17–0.19) $p < 0.0001$ N = 889	0.00 ^a (–0.15–0.13) $p = NS$ N = 889	No data	

BB: beta-blocker; CCB: calcium channel blocker; CI: confidence interval; ETT: exercise tolerance test; LAN: long-acting nitrate; NS: not significant; ^aSingle study used to evaluate this outcome.

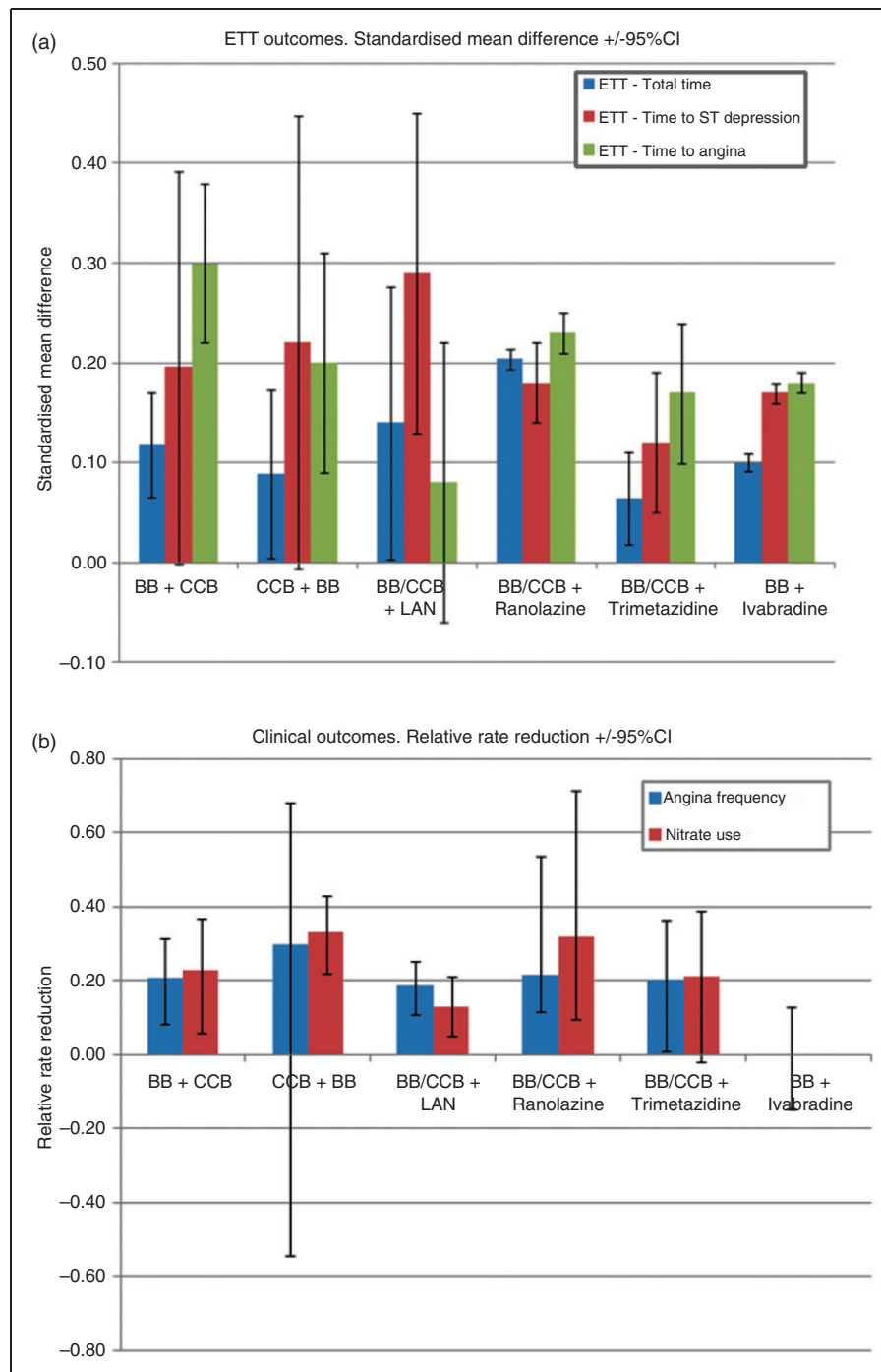


Figure 1. Comparative results across classes: (a) exercise tolerance test (ETT) outcomes and (b) clinical outcomes (angina frequency and nitrate use). BB: beta-blocker; CCB: calcium channel blocker; CI: confidence interval; LAN: long-acting nitrate.

CCB, trimetazidine added to BB or CCB, LAN added to BB or CCB and CCB added to BB. There was no significant difference seen for BB added to CCB, or for ivabradine added to BB. For weekly nitrate usage, statistically significant benefits were seen for ranolazine added to BB or CCB, BB added to CCB and CCB added to BB. There was no significant difference seen

with trimetazidine added to BB or CCB. There was no data for this outcome available for ivabradine.

Subgroup analyses

Detailed results of the subgroup analyses are shown in Table 2. The key findings are summarised below.

Table 2. Subgroup analyses: results for exercise tolerance test (ETT) total time and angina frequency.

Subgroups	ETT total time (SMD; 95% CI; p-value)					Angina frequency (RRR; 95% CI; p-value)				
	BB+CCB	CCB+BB	BB/CCB+LAN	BB/CCB+Ran	BB/CCB+Trim	BB+Ivab	BB+CCB	CCB+BB	BB/CCB+LAN	BB/CCB+Ran
Post-1990	0.065; 0.01–0.12 p = 0.03	–0.004; –0.03–0.03 p = NS	0.156; –0.01–0.32 p = NS	0.204; 0.19–0.21 p < 0.001	0.065; 0.02–0.11 p = 0.006	0.10; ^a 0.09–0.11 p < 0.0001	0.083; –0.11–0.24 p = NS	0.152; –1.25–0.68 p = NS	0.221; 0.12–0.31 p < 0.0001	0.217; 0.10–0.32 p = 0.0006
Pre-1990	0.253; 0.17–0.34 p < 0.0001	0.301; 0.24–0.36 p < 0.0001	0.111; –0.16–0.39 p = NS	n/a	n/a	n/a	0.320; 0.23–0.40 p < 0.0001	0.490; –0.24–0.79 p = NS	0.134; 0.01–0.25 p = 0.04	n/a
≥ 7 attacks per week	0.219; 0.13–0.31 p < 0.0001	0.306; 0.25–0.36 p < 0.0001	0.175; –0.01–0.36 p = NS	n/a	n/a ^b	n/a	0.316; 0.24–0.39 p < 0.0001	0.490; –0.23–0.79 p = NS	0.238; 0.15–0.32 p < 0.0001	n/a
< 7 attacks per week	0.111; 0.03–0.19 p = 0.006	0.048; –0.11–0.21 p = NS	n/a	0.204; 0.19–0.21 p < 0.001	0.083; 0.04–0.13 p = 0.0001	0.10; ^a 0.09–0.11 p < 0.0001	0.065; –0.13–0.23 p = NS	–0.174; –3.56–0.70 p = NS	n/a ^b	0.217; 0.10–0.32 p = 0.0006
Complex features	0.039; –0.05–0.12 p = NS	0.010; –0.01–0.03 p = NS	0.108; –0.06–0.27 p = NS	0.204; 0.19–0.21 p < 0.001	–0.07; –0.04–0.10 p < 0.0001	n/a	0.065; –0.21–0.28 p = NS	0.199; –1.79–0.77 p = NS	0.225; 0.10–0.33 p = 0.00006	0.245; 0.09–0.37 p = 0.003
No complex features	0.157; 0.09–0.23 p < 0.0001	0.083; –0.06–0.23 p = NS	0.257; –0.11–0.62 p = NS	n/a	0.039; 0.01–0.07 p = 0.01	0.10; ^a 0.09–0.11 p < 0.0001	0.282; 0.12–0.42 p = 0.002	0.432; –2.15–0.90 p = NS	0.161; 0.06–0.25 p = 0.002	0.074; ^a –0.06–0.19 p = NS

BB: beta-blocker; CCB: calcium channel blocker; CI: confidence interval; Ivab: ivabradine; NS: not significant; Ran: ranolazine; RRR: relative rate reduction; SMD: standardised mean difference; Trim: trimetazidine; ^aData from single large study; ^bOnly single small study qualifies therefore data not given.

Date of publication

Out of 46 studies included in this analysis, 14 were published before 1990 and 32 after 1990. All studies of ranolazine and ivabradine were published after 1990. For the ETT outcome, earlier studies comparing CCB added to BB and BB added to CCB showed a marked tendency to yield more favourable results compared with the newer studies. The reverse was the case for the angina frequency outcome, with better outcomes tending to be seen in newer studies. For LAN or trimetazidine added to BB or CCB, there was no obvious difference with regard to the publication date, although numbers of studies in each category were small, and therefore had wide confidence intervals.

Baseline angina frequency

A total of 12 studies were carried out in patients with a mean baseline angina frequency of ≥ 7 attacks per week, whereas 21 studies had a mean rate of < 7 per week. The remaining 13 studies did not provide information on angina frequency. All studies investigating ranolazine and ivabradine were carried out in patients with a low baseline attack frequency. A single study investigating trimetazidine was carried out in a high frequency patient group, but as this included only 32 patients per arm, the results were considered unreliable and were not included in the analysis. Similarly, for the combination of BB or CCB+LAN, all studies with ETT outcomes were carried out in populations with a high baseline attack rate, while a single study reporting angina frequency reduction as an outcome was carried out in a low-attack rate group. As this study only had 12 patients per arm, the results were not included in the analysis. For the remaining two treatment groups (BB+CCB and CCB+BB), the magnitude of benefit was greatest in those with a high baseline attack rate.

Patients with advanced disease

Categorising patients according to the severity of coronary disease is difficult, as very limited data are supplied in many of the studies. Of the 46 included studies, 21 supplied information on prior re-vascularisation, 15 on rates of multi-vessel disease and 17 on heart failure status. Twelve studies did not supply data on any of these three variables. The results for this subanalysis should, therefore, be treated with a degree of caution. Overall, for the ETT outcome there was a general tendency for a greater benefit in patients without complex disease, although the small number of studies makes it impossible to comment on ranolazine or ivabradine. For the angina frequency outcome, the results are mixed, with a BB or CCB combination showing the

same tendency towards better outcomes in non-complex patients, while BB or CCB+LAN or ranolazine show the opposite tendency.

Estimation of the effect magnitude

Whilst the clinical outcomes are presented as relative rate reductions, the ETT results are expressed as a SMD, which carries less intuitive meaning. One way to translate this measure into an estimate of effect size is to multiply the result by a typical standard deviation observed in the population under test – this will then yield an expected mean difference in ETT for a typical patient group. By using this figure to generate a percentage improvement in ETT duration, it is possible to gain some understanding of the clinical relevance of the results, albeit subject to the caveats described in the methods section regarding differing exercise protocols.

Applying this methodology to the total exercise time outcome, using the pooled mean and standard deviation derived from control patients, it can be estimated that adding CCB to BB or adding BB to CCB prolonged the total exercise duration by approximately 3.3% and approximately 2.4% respectively. When added to BB or CCB, LAN as well as ranolazine were associated with a 3.9% increase in the total exercise time, whereas adding trimetazidine or ivabradine to BB or CCB resulted in approximately 1.4% and 2.5% increases in this parameter respectively.

Discussion

Main findings

This systematic review and meta-analysis has assessed the impact of a range of anti-anginal classes on both ETT and clinical outcomes, when used as add-on therapy in patients with SCAD and residual angina on either BB or CCB. The results support the commonly practised strategy of initially administering CCB and BB as dual therapy, this combination yielding benefit for most of the comparisons tested.

Beyond this, ranolazine added to either BB or CCB consistently resulted in a clinically and statistically significant improvement in angina control, as assessed by these outcomes (Table 1). Trimetazidine also showed benefit across most outcomes, although considerably greater between-studies heterogeneity was seen. The evidence for LAN used in an add-on role is less clear-cut: although there were statistically significant benefits seen for most of the outcomes tested, the component studies in these results were small and of variable quality, thereby yielding wide confidence intervals around the results. The single study of ivabradine added to BB showed ETT benefits broadly comparable to those seen

with trimetazidine, but no reduction in angina frequency was seen. No study was identified that assessed the effect of nicorandil on angina severity, when used in an add-on role.

Interpretation of the pooled estimates of treatment effect obtained in this analysis requires consideration of the methods used. For the pure clinical outcomes (angina frequency and short-acting nitrate use) the results are presented as a relative rate reduction, which is intuitively straightforward. Thus, the addition of CCB to BB may be expected to yield an 18% reduction in angina frequency, the addition of LAN to BB or CCB a 19% reduction, the addition of ranolazine to CCB or BB a 25% reduction, and the addition of trimetazidine to BB or CCB a 21% reduction (all other combinations yielding no significant benefit). Similar results can be drawn from the data for short acting nitrate use, with the combination of ranolazine and BB or CCB yielding the greatest benefit.

The results for ETT testing are more challenging to interpret, owing to the dimensionless nature of the measure. In principle, however, the larger the SMD the greater the magnitude of treatment effect.

Directly comparing the results for the absolute efficacy between the different treatment options is prone to error, because of differences in study design, patient populations, baseline angina severity and ETT methodology. Although the statistical approach adopted standardises for this, it makes it difficult to make firm comparisons between treatment groups, and the results should be treated with caution. It is probably reasonable to state, however, that where consistent benefits have been demonstrated across both ETT and clinical outcomes there is no clear specific evidence to support the preferential use of one combination over another. Particularly for the combinations of BB+CCB, CCB+BB and BB or CCB+ranolazine, the general magnitude of benefit is broadly comparable.

There is a wide range of treatments and doses used in the included studies. Not every drug and dose included in this analysis will be licensed for use in every country – in order to maximise the relevance of this analysis, a decision was taken not to restrict inclusion based on these factors, but this may have implications for the generalisability of results in any given country.

The results are broadly consistent with the conclusions of Klein et al.,⁴ in their 2002 meta-analysis of 22 BB or CCB combination studies. Based on studies published up to 1999 and using slightly different meta-analytical methodology, they estimated that the combination of BB+CCB was associated with an SMD of 0.15 for total exercise time, with CCB+BB yielding a figure of 0.06. This compares with results of 0.12 and 0.09 from our analysis. For time to 1 mm ST-depression, Klein et al. derived results of 0.26 for

BB+CCB and 0.22 for CCB+BB. Our equivalent results were 0.20 and 0.22 respectively. For time to angina, their results were 0.24 for BB+CCB and 0.13 for CCB+BB, while the equivalents in our analysis were 0.30 and 0.20. Klein et al. did not look at clinical outcomes. The overall impression from this comparison is that, although the conclusions of the two analyses were qualitatively similar, our analysis tended to be slightly more conservative in its estimates of benefit.

Extrapolation of clinical trial results to a clinical context requires a degree of caution. Most studies carried out on anti-anginal agents focus on ETT results, as this represents an objective and reproducible test of efficacy. It is also mandated by European licensing authorities for the evaluation of new agents.⁵³ However, changes in ETT duration are only poorly correlated with changes in patient quality of life,⁵⁴ such that clinical changes associated with attack frequency and severity are a better guide to the impact on quality of life.^{55,56} It has been demonstrated that patients attach considerable value to even quite small changes in the frequency of angina attacks.⁵⁷ This effect is apparent when comparing the clinical and ETT results of our analysis. Whilst improvements in total exercise time ranged from 1–4%, the reductions in angina frequency and nitrate use were generally in the range 20–30%.

Limitations

In common with any systematic review, the results of this analysis reflect the inherent limitations of the methodology. Although every effort has been made to include the totality of published literature on the subject, it is possible that some studies have escaped our attention. Equally, the validity of the meta-analysis depends on the quality of the included studies. Although all were nominally randomised, double-blind controlled trials, the explicit methods used were rarely specified in the published papers, so we cannot be sure of the robustness of the methodology. This reservation is particularly pertinent to the older studies (chiefly comparing BB and CCB combinations), where studies tended to be small and some were of doubtful quality.

A second potential source of error relates to heterogeneity in the recruited patient populations. Although all studies were carried out in patients with symptomatic angina on monotherapy treatment at baseline, the severity of angina varied widely between studies, with treated mean attack rates ranging from <1 to 28 attacks per week. It is possible that the relative efficacy of different treatment choices will vary according to the background severity of angina – however, there were too few studies available to carry out meaningful sub-analyses of this variable.

Within the treatment classes chosen, there is also scope for confounding of results. For some classes – particularly the CCBs – there may be significant between-drugs differences in efficacy. In order to maintain the power of the results, we chose not to disaggregate groups, but once again this may compromise the applicability of the results to a particular situation.

Implications for practice

The use of BB as first-line therapy in the management of stable angina is well established, with other classes of agent tending to be used if BB is contraindicated, ineffective or poorly tolerated. This approach to management reflects extensive evidence for the anti-anginal effectiveness for BBs and many decades of experience in their use. Although there may also be an expectation amongst clinicians that long-term beta blockade will also tend to reduce the risk of adverse cardiovascular outcomes in these patients, recent evidence from both meta-analysis of randomised controlled trials⁵⁸ and prospective registry follow-up⁵⁹ suggests that this is in fact not the case, even for those patients with a prior myocardial infarct.⁵⁹ The decision as to which class of agent should be used – whether for first-line therapy or as second-line add-on – should therefore be made purely on the basis of the likely anti-anginal efficacy and the extent to which the chosen therapy fits with the patient's clinical circumstances.

In this review we have attempted to place data from multiple comparisons into a common context, in order to inform these prescribing decisions. The 2013 European Society of Cardiology (ESC) Guidelines on the management of SCAD² and NICE guidelines¹ recommend BBs and CCBs as first-line antianginal agents used initially alone, and then in combination. The combined effect of these therapies has been extensively documented, usually when CCB has been added to previous therapy with BB, and the efficacy of these combinations has been confirmed by the present meta-analysis.

It should be remembered, however, that individual patient circumstances are likely to be a better indicator than any systematic hierarchy of benefits. The mechanism of action and adverse effect profile of individual drugs usually suggest a particular sequence of therapy to be evaluated in specific patients. This is reflected in the 2013 ESC recommendations which emphasise the importance of individual patient characteristics such as comorbidities and a haemodynamic response when selecting add-on therapy.² The guidelines also offer a possibility of using second-line agents as first-line therapy in selected patients depending on

comorbidities and tolerance, although this recommendation is based on expert consensus rather than the strong evidence. For example, ivabradine may be best indicated for patient with high heart rates that persist despite optimal beta blockade, or in patients for whom beta blockers are contraindicated. On the other hand ranolazine may be specifically required in those patients in whom haemodynamic effects such as bradycardia, hypotension or atrioventricular block must be avoided. Similarly, patients suffering from angina who also have diabetes may be an especially good target for treatment with ranolazine because of favourable results obtained in this subpopulation^{43,60,61} as suggested by recent ESC guidelines.² In the 2012 American College of Cardiology Foundation and the American Heart Association Practice Guidelines, ranolazine has been assigned a Class IIa recommendation for the combined use with BB when initial treatment with BB is not successful (LOE A) or as a first-line substitute for BB if the latter is contraindicated (LOE B).³

Nonetheless the documented efficacy of the drug as an add-on therapy is also a very relevant factor in the choice of antianginal agent. The current recommendations^{1,2} do not give advice about specific add-on therapy combinations, but the present analysis gives a clearer perspective with regard to the balance of improved exercise capacity, reduced frequency of spontaneous angina and need for additional nitrate therapy for the range of second-line therapies added to BB or CCB. These results facilitate an evidence-based approach to selection of add-on therapy.

Conclusion

Across a broad range of commonly assessed outcomes, there is little difference in the magnitude of clinical efficacy of different anti-anginal classes, used in an add-on role. The commonly practised strategy of using a combination of BB and CCB is generally supported by good evidence of symptomatic benefit. Of the other available options, ranolazine added to either BB or CCB offers consistently significant gains, while LAN and trimetazidine may also be effective, although the evidence for these is somewhat more heterogeneous. Ivabradine, although associated with an improvement in ETT in a single study, failed to demonstrate clinical benefits; there are no data for nicorandil in this role.

Funding

This work was supported by a grant from A Menarini Farmaceutica Internazionale.

Conflict of interest

J Belsey reports grants from A Menarini Farmaceutica Internazionale during the conduct of the study; grants and personal fees from Daiichi-Sankyo Europe outside of the study.

J Camm reports personal fees from A Menarini Farmaceutica Internazionale during the conduct of the study; personal fees from Astra Zeneca, ChanRX, Gilead, Merck, Otsuka, Sanofi, Servier, Xention, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Daiichi-Sankyo, Pfizer, Boston Scientific, Biotronik, Medtronic, St Jude Medical, Actelion, GlaxoSmithKline, Infobionic, Incarda, Johnson & Johnson, Mitsubishi, Novartis and Takeda outside the study.

A Mugelli reports personal fees from A Menarini Farmaceutica Internazionale during the conduct of the study; personal fees from Bayer, Chiesi and Novartis outside the study.

I Savelieva reports personal fees from A Menarini Farmaceutica Internazionale during the conduct of the study; personal fees from Boehringer Ingelheim, Pfizer, Bristol Myers Squibb, Bayer, and Sanofi and Servier outside the study.

Acknowledgement

This work has previously been presented as a poster at the American Heart Association Scientific Sessions, Los Angeles, 5 November 2012.

References

1. National Institute for Health and Clinical Excellence (NICE). *CG126 Stable angina*. Manchester: National Clinical Guidelines Centre, 2011.
2. Montalescot G, Sechtem U, Achenbach S, et al. 2013 ESC guidelines on the management of stable coronary artery disease: The Task Force on the management of stable coronary artery disease of the European Society of Cardiology. *Eur Heart J* 2013; 34: 2949–3003.
3. Fihn SD, Gardin JM, Abrams J, et al. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS Guideline for the diagnosis and management of patients with stable ischemic heart disease: A report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, and the American College of Physicians, American Association for Thoracic Surgery, Preventive Cardiovascular Nurses Association, Society for Cardiovascular Angiography and Interventions, and Society of Thoracic Surgeons. *J Am Coll Cardiol* 2012; 60: e44–e164.
4. Klein WW, Jackson G and Tavazzi L. Efficacy of monotherapy compared with combined antianginal drugs in the treatment of chronic stable angina pectoris: A meta-analysis. *Coron Artery Dis* 2002; 13: 427–436.
5. Hozo SP, Djulbegovic B and Hozo I. Estimating the mean and variance from the median, range, and the size of a sample. *BMC Med Res Methodol* 2005; 5: 13.
6. DerSimonian R and Laird N. Meta-analysis in clinical trials. *Control Clin Trials* 1986; 7: 177–188.
7. Akhras F and Jackson G. Efficacy of nifedipine and isosorbide mononitrate in combination with atenolol in stable angina. *Lancet* 1991; 338: 1036–1039.
8. Donaldson KM, Dawkins KD and Waller DG. A comparison of nisoldipine and nifedipine, in combination with atenolol, in the management of myocardial ischaemia. *Eur Heart J* 1993; 14: 534–539.
9. Dunselman P, Liem AH, Verdel G, et al. Addition of felodipine to metoprolol vs replacement of metoprolol by felodipine in patients with angina pectoris despite adequate beta-blockade. Results of the Felodipine ER and Metoprolol CR in Angina (FEMINA) Study. Working Group on Cardiovascular Research, The Netherlands (WCN). *Eur Heart J* 1997; 18: 1755–1764.
10. Dunselman PH, Van Kempen LH, Bouwens LH, et al. Value of the addition of amlodipine to atenolol in patients with angina pectoris despite adequate beta blockade. *Am J Cardiol* 1998; 81: 128–132.
11. El-Tamimi H and Davies GJ. Optimal control of myocardial ischaemia: The benefit of a fixed combination of atenolol and nifedipine in patients with chronic stable angina. *Br Heart J* 1992; 68: 291–295.
12. Emanuelsson H, Egstrup K, Nikus K, et al. Antianginal efficacy of the combination of felodipine-metoprolol 10/100 mg compared with each drug alone in patients with stable effort-induced angina pectoris: A multicenter parallel group study. The TRAFFIC Study Group. *Am Heart J* 1999; 137: 854–862.
13. Foale RA. Atenolol versus the fixed combination of atenolol and nifedipine in stable angina pectoris. *Eur Heart J* 1993; 14: 1369–1374.
14. Fox KM, Mulcahy D, Findlay I, et al. The Total Ischaemic Burden European Trial (TIBET). Effects of atenolol, nifedipine SR and their combination on the exercise test and the total ischaemic burden in 608 patients with stable angina. The TIBET Study Group. *Eur Heart J* 1996; 17: 96–103.
15. Humen DP, O'Brien P, Purves P, et al. Effort angina with adequate beta-receptor blockade: Comparison with diltiazem alone and in combination. *J Am Coll Cardiol* 1986; 7: 329–335.
16. Hung J, Lamb IH, Connolly SJ, et al. The effect of diltiazem and propranolol, alone and in combination, on exercise performance and left ventricular function in patients with stable effort angina: A double-blind, randomized, and placebo-controlled study. *Circulation* 1983; 68: 560–567.
17. Jenkins RM and Nagle RE. The symptomatic and objective effects of nifedipine in combination with beta-blocker therapy in severe angina pectoris. *Postgrad Med J* 1982; 58: 697–700.
18. Johnston DL, Lesoway R, Humen DP, et al. Clinical and hemodynamic evaluation of propranolol in combination with verapamil, nifedipine and diltiazem in exertional angina pectoris: A placebo-controlled, double-blind, randomized, crossover study. *Am J Cardiol* 1985; 55: 680–687.
19. Kawanishi DT, Reid CL, Morrison EC, et al. Response of angina and ischemia to long-term treatment in patients with chronic stable angina: A double-blind randomized

- individualized dosing trial of nifedipine, propranolol and their combination. *J Am Coll Cardiol* 1992; 19: 409–417.
20. Kenny J, Kiff P, Holmes J, et al. Beneficial effects of diltiazem and propranolol, alone and in combination, in patients with stable angina pectoris. *Br Heart J* 1985; 53: 43–46.
 21. Madjlessi-Simon T, Fillette F, Mary-Krause M, et al. Effects of amlodipine on transient myocardial ischemia in patients with a severe coronary condition treated with a beta-blocker. *Eur Heart J* 1995; 16: 1780–1788.
 22. Meyer TE, Adnams C and Commerford P. Comparison of the efficacy of atenolol and its combination with slow-release nifedipine in chronic stable angina. *Cardiovasc Drugs Ther* 1993; 7: 909–913.
 23. O'Keefe JC, Creamer JE and Banim SO. Efficacy of nisoldipine combined with beta-adrenergic-blocking drugs in the treatment of chronic stable angina. *Clin Cardiol* 1987; 10: 345–350.
 24. Parameshwar J, Keegan J, Mulcahy D, et al. Atenolol or nicardipine alone is as efficacious in stable angina as their combination: A double blind randomised trial. *Int J Cardiol* 1993; 40: 135–141.
 25. Pehrsson SK, Ringqvist I, Ekdahl S, et al. Monotherapy with amlodipine or atenolol versus their combination in stable angina pectoris. *Clin Cardiol* 2000; 23: 763–770.
 26. Ronnevik PK, Silke B and Ostergaard O. Felodipine in addition to beta-adrenergic blockade for angina pectoris. a multicentre, randomized, placebo-controlled trial. *Eur Heart J* 1995; 16: 1535–1541.
 27. Rossinen J, Partanen J and Nieminen MS. Amlodipine in patients with stable angina pectoris treated with beta-blockers. Double-blind comparison with placebo. *Scand Cardiovasc J* 1998; 32: 41–48.
 28. Sangiorgio P, Di PG, Savonitto S, et al. Felodipine in chronic stable angina: A randomized, double-blind, placebo-controlled, crossover study. *Eur Heart J* 1990; 11: 1011–1017.
 29. Savonitto S, Ardissio D, Egstrup K, et al. Combination therapy with metoprolol and nifedipine versus monotherapy in patients with stable angina pectoris. Results of the International Multicenter Angina Exercise (IMAGE) Study. *J Am Coll Cardiol* 1996; 27: 311–316.
 30. Strauss WE and Parisi AF. Randomized, placebo-controlled evaluation of lidoflazine in patients receiving beta-blocker therapy with propranolol: Improvement in anginal symptoms and exercise tolerance with combined drug therapy. *Am J Cardiol* 1984; 53: 1003–1007.
 31. Strauss WE and Parisi AF. Superiority of combined diltiazem and propranolol therapy for angina pectoris. *Circulation* 1985; 71: 951–957.
 32. Toal CB, Motro M, Baird MG, et al. Effectiveness of nifedipine GITS in combination with atenolol in chronic stable angina. *Can J Cardiol* 1999; 15: 1103–1109.
 33. Tweddel AC, Beattie JM, Murray RG, et al. The combination of nifedipine and propranolol in the management of patients with angina pectoris. *Br J Clin Pharmacol* 1981; 12: 229–233.
 34. Uusitalo A, Arstila M, Bae EA, et al. Metoprolol, nifedipine, and the combination in stable effort angina pectoris. *Am J Cardiol* 1986; 57: 733–737.
 35. Woodmansey PA, Stewart AG, Morice AH, et al. Amlodipine in patients with angina uncontrolled by atenolol. A double blind placebo controlled cross over trial. *Eur J Clin Pharmacol* 1993; 45: 107–111.
 36. Humen DP and Kostuk WJ. Clinical response and effects on left ventricular function of isosorbide dinitrate added to propranolol or diltiazem monotherapy in patients with chronic stable angina. *Can J Cardiol* 1991; 7: 74–80.
 37. Jensen G, Trautner F, Rasmussen S, et al. Haemodynamic and clinical effects of isosorbide-dinitrate in patients with severe effort angina on beta-blocking treatment. *Eur J Clin Pharmacol* 1983; 24: 169–172.
 38. Svendsen JH, Aldershvile J, Abildgaard U, et al. Efficacy of controlled-release isosorbide-5-mononitrate as adjunctive treatment to beta-blocking agents in patients with stable angina pectoris. *J Cardiovasc Pharmacol* 1989; 14: 358–363.
 39. Friedensohn A, Meshulam R and Schlesinger Z. Randomized double-blind comparison of the effects of isosorbide dinitrate retard, verapamil sustained-release, and their combination on myocardial ischemic episodes. *Cardiology* 1991; 79: S31–S40.
 40. Vlay SC and Olson LC. Nifedipine and isosorbide dinitrate alone and in combination for patients with chronic stable angina: A double-blind crossover study. *Am Heart J* 1990; 120: 303–307.
 41. Chaitman BR, Pepine CJ, Parker JO, et al. Effects of ranolazine with atenolol, amlodipine, or diltiazem on exercise tolerance and angina frequency in patients with severe chronic angina: A randomized controlled trial. *JAMA* 2004; 291: 309–316.
 42. Stone PH, Gratsiansky NA, Blokhin A, et al. Antianginal efficacy of ranolazine when added to treatment with amlodipine: The ERICA (Efficacy of Ranolazine in Chronic Angina) trial. *J Am Coll Cardiol* 2006; 48: 566–575.
 43. Kosiborod M, Arnold SV, Spertus JA, et al. Evaluation of ranolazine in patients with type 2 diabetes mellitus and chronic stable angina: Results from the TERISA randomized clinical trial (Type 2 Diabetes Evaluation of Ranolazine in Subjects With Chronic Stable Angina). *J Am Coll Cardiol* 2013; 61: 2038–2045.
 44. Brochier M, Demange J, Ducloux G, et al. Value of the combination of trimetazidine and a calcium inhibitor in the treatment of chronic coronary insufficiency. Double-blind controlled study versus placebo. *Ann Cardiol Angeiol (Paris)* 1986; 35: 49–56.
 45. Chazov EI, Lepakchin VK, Zharova EA, et al. Trimetazidine in Angina Combination Therapy—the TACT study: Trimetazidine versus conventional treatment in patients with stable angina pectoris in a randomized, placebo-controlled, multicenter study. *Am J Ther* 2005; 12: 35–42.
 46. Chutz DS. Value of addition of trimetazidine to calcium antagonist treatment in chronic coronary insufficiency. *Gaz Med (Paris)* 1984; 91: S93–S95.

47. Danchin N, Marzilli M, Parkhomenko A, et al. Efficacy comparison of trimetazidine with therapeutic alternatives in stable angina pectoris: A network meta-analysis. *Cardiology* 2011; 120: 59–72.
48. Levy S. Combination therapy of trimetazidine with diltiazem in patients with coronary artery disease. Group of South of France Investigators. *Am J Cardiol* 1995; 76: 12B–16B.
49. Manchanda SC and Krishnaswami S. Combination treatment with trimetazidine and diltiazem in stable angina pectoris. *Heart* 1997; 78: 353–357.
50. Sellier P and Broustet JP. Assessment of anti-ischemic and antianginal effect at trough plasma concentration and safety of trimetazidine MR 35 mg in patients with stable angina pectoris: A multicenter, double-blind, placebo-controlled study. *Am J Cardiovasc Drugs* 2003; 3: 361–369.
51. Szwed H, Sadowski Z, Elikowski W, et al. Combination treatment in stable effort angina using trimetazidine and metoprolol: Results of a randomized, double-blind, multicentre study (TRIMPOL II). TRIMetazidine in POLand. *Eur Heart J* 2001; 22: 2267–2274.
52. Tardif JC, Ponikowski P and Kahan T. Efficacy of the I(f) current inhibitor ivabradine in patients with chronic stable angina receiving beta-blocker therapy: A 4-month, randomized, placebo-controlled trial. *Eur Heart J* 2009; 30: 540–548.
53. The Committee for Medicinal Products for Human Use (CHMP). *Guideline on the clinical investigation of anti-anginal medicinal products in stable angina pectoris*. CPMP/EWP/234/95/rev.1. London: European Medicines Agency, June 2006.
54. Wiklund I, Comerford MB and Dimenas E. The relationship between exercise tolerance and quality of life in angina pectoris. *Clin Cardiol* 1991; 14: 204–208.
55. Longworth L, Buxton MJ, Sculpher M, et al. Estimating utility data from clinical indicators for patients with stable angina. *Eur J Health Econ* 2005; 6: 347–353.
56. Goldsmith KA, Dyer MT, Buxton MJ, et al. Mapping of the EQ-5D index from clinical outcome measures and demographic variables in patients with coronary heart disease. *Health Qual Life Outcomes* 2010; 8: 54.
57. Chestnut LG, Keller LR, Lambert WE, et al. Measuring heart patients' willingness to pay for changes in angina symptoms. *Med Decis Making* 1996; 16: 65–77.
58. Shu de F, Dong BR, Lin XF, et al. Long-term beta blockers for stable angina: Systematic review and meta-analysis. *Eur J Prev Cardiol* 2012; 19: 330–341.
59. Bangalore S, Steg G, Deedwania P, et al.; REACH Registry Investigators. Beta-blocker use and clinical outcomes in stable outpatients with and without coronary artery disease. *JAMA* 2012; 308: 1340–1349.
60. Timmis AD, Chaitman BR and Crager M. Effects of ranolazine on exercise tolerance and HbA1c in patients with chronic angina and diabetes. *Eur Heart J* 2006; 27: 42–48.
61. Chisholm JW, Goldfine AB, Dhalla AK, et al. Effect of ranolazine on A1C and glucose levels in hyperglycemic patients with non-ST elevation acute coronary syndrome. *Diabetes Care* 2010; 33: 1163–1168.